American Computer Science League

2025-2026 ● Contest 2: Short Problems ● Junior Division

1. Prefix/Infix/Postfix Notation

Convert the following infix expression to prefix:

$8 - 3 * 2 + 2^3 - 8 / 2$

A. − + − * ^ / 8 3 2 2 3 8 2
B. − − + 8 * 3 2 ^ 2 3 / 8 2
C. − + − 8 * 3 2 ^ 2 3 / 8 2
D. − + − 8 * 2 3 ^ 3 2 / 2 8
E. − + − 8 ^ 3 2 * 2 3 / 8 2

2. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if a = 3, b = -2, and c = 5:

a c + a b − * a b − c + / 2 ^

A. $\frac{2}{5}$
B. 1
C. $\frac{16}{9}$
D. 16
E. -16

3. Bit-String Flicking

Evaluate:

NOT (10110 OR 11100 AND NOT 01101)

A. 10001
B. 00101
C. 00011
D. 10100
E. 01001

4. Bit-String Flicking

Evaluate:

(RCIRC-3 ((LCIRC-2 01101) AND (RSHIFT-2 11010)))

A. 01100
B. 10000
C. 00100
D. 10001
E. 01010

5. What Does This Program Do? (Loops)

What is the final value of count when the following program is run?

```
count = 0
for x = 1 to 9
     num = x * 100 + x * 10 + x
     sum = 3 * x
     if num / sum == 37 then
          count = count + 1
     end if
next x
```

A. 0
B. 1
C. 3
D. 5
E. 9

6. What Does This Program Do? (Loops)

What is output by the following program?

```
num = 1089
for x = 3 to 11
     count = 0
     while num % x == 0
          count = count + 1
          num = num / x
     end while
     if count != 0 then
          output x, count
     end if
next x
```

A. 3 3
   11 2
B. 3 2
   11 2
C. 3 2
   9 1
   11 2
D. 3 1
   3 1
   11 1
   11 1
E. 11 2
   3 2

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1. 前缀/中缀/后缀表示法

将以下中缀表达式转换为前缀表达式：

$8 - 3 * 2 + 2^3 - 8 / 2$

A. − + − * ^ / 8 3 2 2 3 8 2
B. − − + 8 * 3 2 ^ 2 3 / 8 2
C. − + − 8 * 3 2 ^ 2 3 / 8 2
D. − + − 8 * 2 3 ^ 3 2 / 2 8
E. − + − 8 ^ 3 2 * 2 3 / 8 2

2. 前缀/中缀/后缀表示法

当 a = 3, b = -2, 且 c = 5 时，对以下后缀表达式进行求值：

a c + a b − * a b − c + / 2 ^

A. $\frac{2}{5}$
B. 1
C. $\frac{16}{9}$
D. 16
E. -16

3. 位运算

计算：

NOT (10110 OR 11100 AND NOT 01101)

A. 10001
B. 00101
C. 00011
D. 10100
E. 01001

4. 位运算

计算：

(RCIRC-3 ((LCIRC-2 01101) AND (RSHIFT-2 11010)))

A. 01100
B. 10000
C. 00100
D. 10001
E. 01010

5. 这个程序是做什么的？(循环)

运行下列程序时，变量 count 的最终值是多少？

```
count = 0
for x = 1 to 9
     num = x * 100 + x * 10 + x
     sum = 3 * x
     if num / sum == 37 then
          count = count + 1
     end if
next x
```

A. 0
B. 1
C. 3
D. 5
E. 9

6. 这个程序是做什么的？(循环)

下列程序的输出结果是什么？

```
num = 1089
for x = 3 to 11
     count = 0
     while num % x == 0
          count = count + 1
          num = num / x
     end while
     if count != 0 then
          output x, count
     end if
next x
```

A.
```
3 3
11 2
```
B.
```
3 2
11 2
```
C.
```
3 2
9 1
11 2
```
D.
```
3 1
3 1
11 1
11 1
```
E.
```
11 2
3 2
```